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Branding is
what people
say about
you when
you are not
in the room.



-Jeff Bezos

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reduce altitude, while undersized fins may not provide enough stabilisation force.

Fin placement is also very important. Fins are generally mounted at the extreme rear of the rocket because placing them farther back increases stability by shifting the centre of pressure rearward. Even small alignment errors in fins can create unwanted rotational torque, causing the rocket to spiral or drift during flight.

The thickness of the fins must also be chosen carefully. Thin fins reduce drag but may bend or break under aerodynamic loads. Thick fins improve strength but add weight and drag. For most model rockets, fin thicknesses between 2mm and 5mm work well, depending on rocket size and material.

Servo control of the fin. In servo-controlled fin systems, the flight controller continuously reads data from sensors such as accelerometers and gyroscopes and adjusts the fin angles to correct the rocket's orientation. One important rule in servo fin control is limiting the fin deflection angle. Large fin angles create excessive drag and instability. Most active control systems keep fin movement within:

Fin deflection angle $\approx \pm 5^\circ$ to $\pm 15^\circ$

Rocket engine. The rocket engine is the propulsion system that generates thrust to lift the rocket against gravity. It works on Newton's Third Law, where high-speed gases are expelled from the nozzle to produce a force in the opposite direction.

The thrust generated by the

engine is given by:

$$F = \dot{m}V_e + (P_e - P_a)A_e$$

where $\dot{m}$ is the mass flow rate of exhaust gases, V_e is the exhaust velocity, P_e is the exhaust pressure, P_a is the atmospheric pressure, and A_e is the nozzle exit area.

One of the most important rules while selecting a rocket motor is maintaining a proper thrust-to-weight ratio:

Simulation

The next step is to simulate the rocket and analyse the output to determine whether any fine-tuning is required in the design. From the simulation graphs, important parameters such as altitude, velocity, acceleration, stage ignition, recovery deployment, and overall rocket stability can be studied. The altitude graph shows the maximum height reached, while the velocity and acceleration graphs help analyse motor performance and flight behaviour. Stability analysis verifies proper CG and CP positioning for safe flight.

After satisfactory simulation results are obtained, the rocket parts can be assembled, and the control and telemetry code can be uploaded to the IndusBoard Coin for testing and flight operation. Fig. 4 shows OpenRocket flight simulation result displaying vertical motion versus time for the model rocket using a clustered A6-4 motor configuration. The graph displays altitude, vertical velocity, and vertical acceleration during launch, coast phase, apogee, recovery deployment, and descent. The simulation predicts a maximum altitude of approximately 112m with a peak velocity of around 43.9m/s.

EFY note. This is the first part covering rocket design and simulation. The next part will cover the assembly process, circuit connections, flight computer coding, and testing procedure.

To be continued...

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